

MIE366 – DESIGN ASSIGNMENT 1: “DESIGN PROPOSAL”

Department of Mechanical and Industrial Engineering – University of Toronto

[DUE DATE: September 21st, end-of-day, Toronto time]

[GROUP SIZE: 2]

[WEIGHT: 4%]

[ONLINE SUBMISSION]

Learning Objectives

In the labs, you will see and build a variety of circuits which span all new components and analysis techniques covered in the lectures. However, these labs are designed to guide you towards specific solutions and key results. As a Mechatronics engineer, you also need to be able to devise solutions to open-ended problems, and see them through from an engineering specification to final design. As such, the key learning objectives for the design assignment series are:

1. Learn how to select and apply appropriate circuits and circuit design patterns to solve an open-ended electronics problem, including research of existing solutions.
2. Apply simulation tools from the labs (LTSPICE) and analysis techniques from the lectures to design a custom solution to the problem at hand.
3. Create professional prototypes of a circuit using appropriate building techniques. This includes being able to *design professional printed circuit boards (PCBs)* and to *solder through-hole components*.
4. Build, test, and validate prototypes of circuits against a specification, and make modifications to address real-world differences in circuit operation versus simulated behaviour.

To this end, across four design assignments you will complete the engineering design process for a moderately complex circuit design problem. You will create a preliminary set of candidate designs, prototype one on a PCB, modify the design and create a mechanical enclosure, and create a final set of manufacturing-ready plans and a finished prototype device. In the process we will also teach you hard skills in electronic design, such as how to best use circuit simulation packages and modern design tools, how to solder, and how to layout and prepared PCBs for manufacturing.

If you have not done so already, please read the ‘Design Assignments Roadmap’ document now.

Introduction – Adjustable DC Power Supply

In this and other electronics courses you use the bench-top DC laboratory power supplies to provide stable, clean, and controllable power to the circuits you are building and testing. In fact, all electronic devices you use have some form of power supply in them. The initial source of energy for input to a power supply may be the electrical grid or an energy storage device (battery).

At their core, power supplies modify incoming current to change its form (e.g., converting AC to DC) and to produce a predictable output at suitable voltage and current capacity for the intended load. Most power supplies for consumer products output a fixed voltage and have a fixed current limit, if any. Thus, while suitable for powering their intended load/device, they have limited utility as ‘test equipment’ for powering an arbitrary device-under-test.

Rather, one would use a purpose-designed lab power supply. In this series of design assignments, your goal will be to design and build a low-cost adjustable DC ‘lab’ supply which you can keep and use after the course ends. If one were to write a very general Engineering Specification for such a lab supply, a basic feature set might be:

- Output voltage must be adjustable over a selected range.
- Output current should be able to be limited to a user-selectable value.
- The user should be able to easily set the target output voltage and current limit, and to view the present voltage and current at the output.
- The supply should attempt to maintain its output voltage at the target level set by the user, even when the applied load changes (assuming the current limit is not reached).
- If the current limit is reached, the supply should reduce its output voltage until the output current limit is not exceeded (called ‘*fold-back current limiting*’).

To help us better define the problem, we can also think about the intended use-case. This project is intended to provide you with an entry-level design at low cost; we will not reach the quality of a \$200-\$300 professional supply, but the result should still be usable for many testing scenarios. The design should be able to power common circuits and other loads you may encounter in the undergraduate program – this would include microcontrollers (requiring low noise at the output), motors and mechanical elements (requiring high current), and op-amps, sensors, etc. (requiring a modest voltage range). Based on our experience, we have set the following targets for the overall design:

- Output voltage should be adjustable from $0 - 15\text{ V}$.
 - 0 V minimum ensures you can directly control devices like motors down to zero speed.
 - 15 V maximum is enough to power most analog and digital circuits you encounter, as well as common 12V electromechanical devices like motors and actuators.
- Output current should have an adjustable limit from $0 - 3\text{ A}$.
 - 3 A is sufficient to run modestly sized motors and actuators without overly complicating the design.
- Supply should have no worse than 2% load/line regulation
 - We will discuss these figures in more detail in Lab 4 – basically, we want the output voltage to stay within 2% of target under the worse-case load and input conditions.
- Supply should ideally exhibit no more than 250 mV of noise with worse-case loading.
- Supply should have an isolated output.
 - This means no direct connection to earth ground; it would allow one to connect together two of these supplies to create a $\pm$ voltage as we do with the supplies in the lab.
- Supply should be inherently safe to build, test, and use.
- Supply should be able to accept 19 VDC , 3.42 A as its input power source.

On the last three points, to ensure the device you build is inherently safe and isolated, we do not want you working directly with AC line voltages ($120 - 240\text{ VAC}$, depending on the country) as one might in a consumer design. Instead, we will employ a pre-packaged power “brick” to initially step down the AC line voltage to an inherently safe DC voltage before it enters your design. Our power supply design will thus focus on adding all the features an off-the-shelf fixed-output supply will not have: voltage adjustment, current limiting, and a user interface. In case you are concerned this is not reflective of a real design, consider that most complex devices (such as a laptop) use exactly this approach: a single-voltage external power “brick” or “charger” plus an internal supply to produce multiple voltages.

In fact, we have chosen a laptop adapter as the means of stepping down and converting the line-level AC voltage into a stable DC input for your design. In a more traditional power supply design, the switching regulator/controller section would be provided with high-voltage DC created by directly rectifying the incoming AC line voltage (in other words, 200+ volts DC). Such voltages would represent a significant safety hazard to you while building and testing your circuit, and would make any failure or design error catastrophic. It would also require us to obtain CSA approval for every completed design. An enclosed laptop adapter handles all of these aspects for us and comes pre-approved for plug-in use. Since we will be designing our supply for low-voltage input, you will also be able to test your design with the lab bench supplies in MC402 before connecting the laptop adapter (which is itself *not* current regulated).

An input design point of 19V, 3.42A was specifically chosen to correspond to the most common output rating for surplus laptop adapters. Although we cannot provide one with your kit of parts, if you wish to use your design after the course concludes or otherwise outside of MC402 it should be trivial to find one at reasonable cost (at the time of writing, these could be found on sites like Amazon for around \$15-\$20 or slightly more at local computer stores. In fact, you can use most 19V adapters, even those with lower current ratings, provided you limit the total input power to your design via the firmware we will provide.

Thus, we can now present a block diagram of the power supply design as a whole, and also narrow down exactly what *you* will be creating in this first design assignment.

As you can see from Figure 1 on the following page, we have limited the scope of the design so that you do not have to create the entire supply from scratch. In particular, there are elements of embedded systems design, including analog-to-digital plus digital-to-analog conversion and control loop design, which are beyond the scope of this course. If you find you are interested in these topics beyond what we cover in this assignment, please contact the instructor – there are resources easily available for you to delve more deeply into these aspects of the design. You will also want to consider taking related courses in fourth year, such as MIE444 (Mechatronics Principles) and MIE438 (Embedded Systems) if you find such topics interesting. Again, we are happy to offer advice in this regard.

For this series of assignments, there are a number of sub-problems which will require your design effort. Notably, you will be responsible for designing most of the current measurement and limiting system, and for designing the overall PCB layout, as well as a number of other smaller sub-circuits.

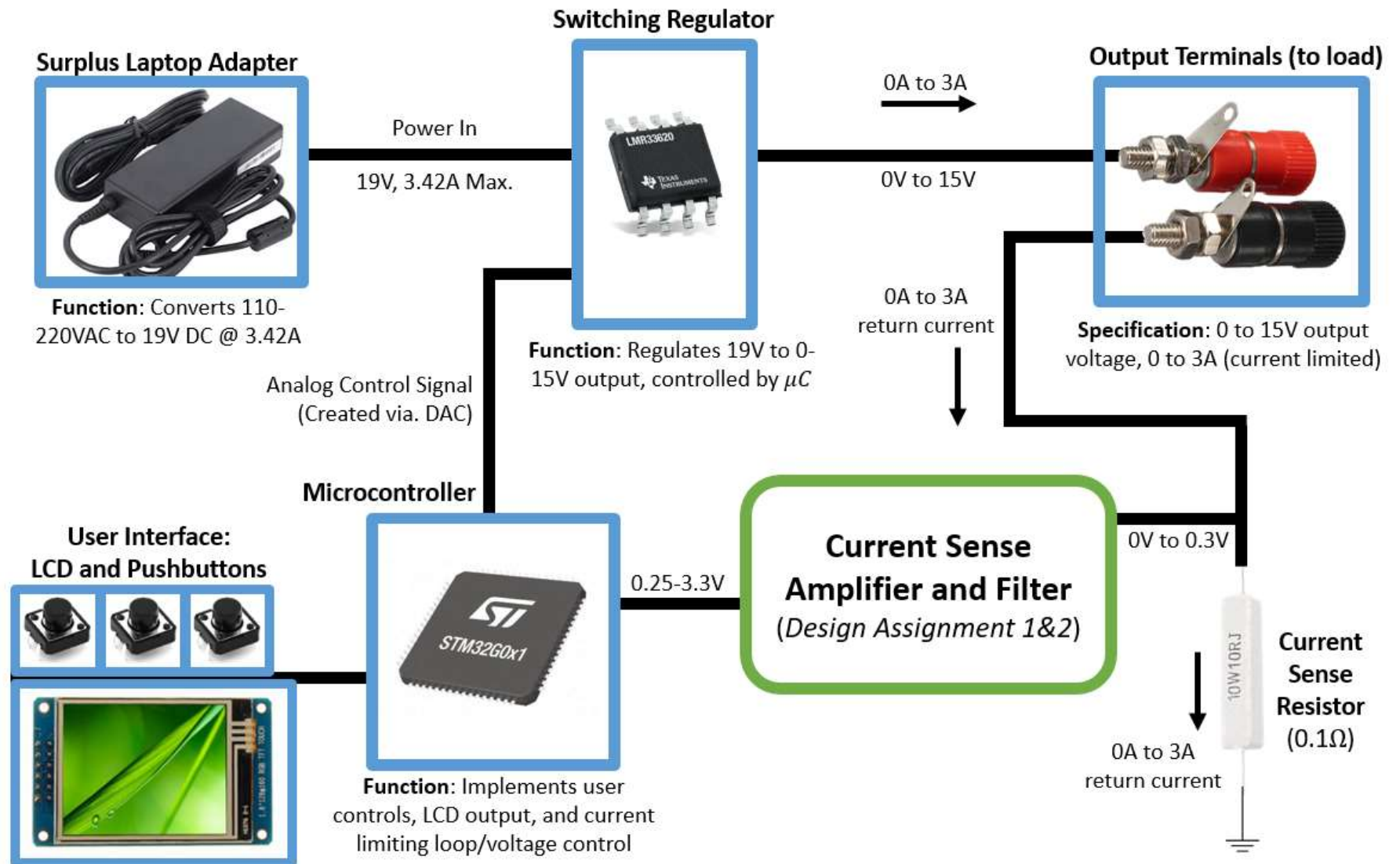


Figure 1 – Visual block diagram of the overall power supply design.

General Design Explanation

A high level explanation of the design in Figure 1 is as follows. Note that we will go into more detail for certain sub-circuits in later design assignments.

- Input power is supplied to your design from a surplus laptop adapter at $19V$, $3.42 A$. This gives us $(19)(3.42) = 65 W$ of input power to work with. At our maximum output, $P_{OUT} = (15)(3) = 45 W$. This means the overall design will need to be at least $\eta = \frac{45 W}{65 W} \times 100\% = 69\%$ efficient in converting input power to output power if we are to achieve the maximum output specification.
- A LMR33630 switching regulator is used to create the primary output voltage. A datasheet for this regulator is included with the assignment posting; you may want to look at it for interest – our use-case closely follows the sample application schematic on Page 19.

In class you will learn about *linear* regulators. These devices produce a regulated DC output by ‘burning off’ the difference from input voltage to desired output voltage as waste-heat. While effective in producing regulation and simple to design, these devices are not efficient. A *switching* regulator is more complex, but more efficient.

As the name implies, this regulator type uses a switching element (transistor) to create *pulses* of DC at a high frequency. An inductor and diode (a new device we will study immediately after filters) are used to convert what is effectively high-frequency AC (pulses) back to DC very efficiently. The output voltage can be varied by changing the ratio of time spent ON versus OFF (called *duty cycle*) for each pulse. This overall process is called PWM, or *pulse-width modulation*. The regulator can also use a *feedback loop* to sense if the output voltage is above or below its target level and automatically adjust the duty cycle to compensate – this achieves a *regulated* output (versus an uncontrolled/unregulated output which would be subject to possibly substantial changes when load is applied). This PWM process itself is also very efficient, meaning the overall implementation wastes little energy if designed correctly. We will review a more detailed explanation of this circuit stage once we have covered diodes and transistors.

- To create a *current-limited* supply, we require two elements, one of which you will design in this assignment:
 - The first element is a controller. The *STM32G051* (a microcontroller, μC) serves this role in our design. The μC will first be used to implement a user interface consisting of some buttons (‘SET’ and ‘DONE’), a rotary encoder (adjust UP/DOWN), and a graphical LCD (to display present V/I and V/I limits, plus overall status). All of these elements are interfaced digitally, meaning we will be providing the firmware for the μC and associated circuit diagrams for the interface. Minimal programming will be necessary on your end, unless you want to add to the firmware or make changes on your own (in which case we are happy to provide support).
 - The controller must act based on sensed current and voltage. Sensing voltage will be implemented in a later design assignment, as it is relatively simple – a simple two-resistor voltage divider at the output can be used to convert the $0 - 15V$ range to a $0 - 3.3V$ range, which can then be accepted by one of the μC analog inputs. However, current sensing is more complex, as we will now discuss.

End of background and introduction;
assignment deliverable details follow in separate document.